Supplementary Information

Supplementary Tables

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^*, where $\beta^* \approx 2.94$ was identified via entropy inflection in the 18-dimensional PCA space (K=23 stored patterns). Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity. Med MRE = median relative error of per-feature means; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	Med MRE (%)
0.1	0.29	0.261	0.538	0.542	0.7
0.3	0.88	0.151	0.550	0.536	0.7
0.5	1.47	0.117	0.562	0.528	0.7
1.0	**2.94**	**0.082**	**0.589**	**0.502**	**0.6**
1.5	4.41	0.067	0.613	0.474	0.7
2.0	5.88	0.058	0.638	0.446	0.7
3.0	8.82	0.048	0.689	0.403	0.8

Table S2: **Multiplicity weighting parameters for conditional generation.** The multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio quantifying the effective number of patterns contributing to generation.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Uncomplicated	18	5	1.11	0.80	23.0
PCOS	3	20	26.67	0.80	4.6
Developed PE	5	18	14.40	0.80	7.7

Table S3: **SA generation hyperparameters.** The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.01	Langevin step size (see Table S8)
T	2,000	Langevin iterations per sample
β^*	2.94	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S4: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients (n=23) using bounded Nelder–Mead optimization in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features under TF-only conditions. Bounds: $\pm\log(100)$. Convergence: $f_{\mathrm{tol}} = 10^{-6}$, 500 iterations, 11 random restarts.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	0.35×
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	0.021×
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	15.5×
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	2.17×
mIIa conversion k	mIIa→IIa	2.3×10^{8}	1.15×10^{9}	5.01×

Units for mIIa conversion: $\mathrm{M}^{-1}\mathrm{s}^{-1}$

Table S5: **Summary of mean relative errors across all 72 features.** Per-visit distribution of MREs between real and SA-generated synthetic patient means. The full feature-by-feature table (72 features × 3 visits = 216 entries) is provided in the code repository as `paper_summary_statistics.csv`.

Visit	Median	Mean	25th pct.	75th pct.	Maximum	MRE below 5%
V1 (baseline)	0.022	0.029	0.008	0.039	0.158	59/72 (81.9%)
V2 (1st tri)	0.009	0.021	0.004	0.022	0.224	65/72 (90.3%)
V3 (3rd tri)	0.011	0.016	0.005	0.024	0.062	69/72 (95.8%)
Pooled (all 3 visits)	**0.012**	**0.022**	**0.005**	**0.028**	**0.224**	**193/216 (89.4%)**

Pooled median MRE 95% bootstrap CI: (0.98%, 1.60%) over 10,000 resamples (seed 42).

Table S6: **Comparison with deep generative baselines.** CTGAN and TVAE were trained on the same 90 per-visit records (23 patients × 3 visits, 72 features) at three epoch counts. SA and MVN results from the main text are shown for reference. CTGAN fails at this sample size (median MRE ≈ 19%), consistent with GAN mode collapse on small datasets. TVAE achieves comparable marginal fidelity at high epoch counts but cannot capture cross-visit covariance or perform conditional generation.

Method	Epochs	Med MRE	Med KS	Corr MAE
SA	–	0.019	0.169	0.200
MVN	–	0.024	0.161	0.090
CTGAN	300	0.189	0.364	0.261
CTGAN	1,000	0.195	0.349	0.265
CTGAN	3,000	0.187	0.341	0.180
TVAE	300	0.037	0.174	0.152
TVAE	1,000	0.026	0.248	0.082
TVAE	3,000	0.018	0.224	0.092

Table S7: **PCA variance threshold sensitivity analysis.** SA generation quality is stable across thresholds from 85% to 99%. Median MRE remains between 1.0% and 1.7% across all thresholds. The 95% threshold used in the main text (bold) is not a critical design choice.

Threshold	d_{PCA}	K/d	β^*	Novelty	Med MRE	Corr MAE
85%	13	1.77	2.94	0.420	0.0098	0.124
90%	15	1.53	2.94	0.477	0.0160	0.126
95%	**18**	**1.28**	**2.94**	**0.500**	**0.0124**	**0.143**
97.5%	20	1.15	2.94	0.543	0.0120	0.135
99%	21	1.10	2.94	0.541	0.0172	0.138

Table S8: **ULA vs MALA sampling diagnostics.** Ten chains of 5,000 iterations each were run with both the Unadjusted Langevin Algorithm (ULA) and the Metropolis-Adjusted Langevin Algorithm (MALA) at step size $\alpha = 0.01$ and $\beta^* = 2.94$ on the 18-dimensional PCA memory (K=23 patterns). The MALA acceptance rate of 99.9% confirms that virtually every ULA proposal satisfies detailed balance, and the indistinguishable energy and effective sample size statistics confirm negligible discretization bias. This comparison bounded the discretization error of the reported sampler. The direct reference for the target itself was the analytic sampler of Eq. (3), introduced below.

Metric	ULA	MALA
Acceptance rate (%)	– (no rejection)	99.9 ± 0.03
Mean energy (post-burn-in)	1.932 ± 0.141	1.886 ± 0.231
τ_{int}	109.5 ± 49.5	106.7 ± 30.4
Effective sample size	41.8 ± 14.2	40.0 ± 10.2

Burn-in: 1,000 iterations discarded. τ_{int}: integrated autocorrelation time estimated with a windowed estimator (threshold 0.05). ESS = $(T - T_{\text{burn}})/\tau_{\text{int}}$.

Table S9: **Memorization diagnostics.** Two complementary checks that SA-generated patients are not memorized copies of the K=23 stored profiles. Novelty score is the angular distance between a synthetic sample and its nearest stored pattern at $\beta = \beta^*$. Nearest-neighbor distances are computed in the per-visit standardized 72-dimensional feature space.

Metric	Value
Mean novelty score, $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$	0.50
Median synth–real nearest-neighbor distance (std. units)	6.14
Median real–real nearest-neighbor distance (std. units)	6.87
Distance ratio (synth–real / real–real)	0.89

Table S10: **Descriptive bootstrap Mann–Whitney diagnostic per feature and condition.** For each of 24 feature–condition pairs, we subsampled the synthetic cohort to match the real sample size, applied a Mann–Whitney U test, and repeated 1,000 times. Frac. $p > 0.05$ is the fraction of replicates in which the test detected no difference at the available sample size. This power-limited diagnostic is not an equivalence test. 20 of 24 pairs achieve $p > 0.05$ in $\geq$ 90% of replicates; the four failing pairs (italicized) all involve high-variance features in the smallest subgroups.

Condition	Feature	MRE	Frac. $p > 0.05$
Uncomplicated (n=18)	FVIII	0.057	0.952
Uncomplicated (n=18)	vWF	0.022	0.999
Uncomplicated (n=18)	FIX	0.018	0.998
Uncomplicated (n=18)	FV	0.006	0.996
Uncomplicated (n=18)	α_2-AP	0.019	1.000
Uncomplicated (n=18)	FVII	0.057	0.941
Uncomplicated (n=18)	Fbgn	0.017	0.987
Uncomplicated (n=18)	Plgn	0.020	0.988
PCOS (n=3)	FVIII	0.109	0.985
PCOS (n=3)	vWF	0.150	0.959
PCOS (n=3)	*FIX*	*0.121*	*0.873*
PCOS (n=3)	FV	0.057	0.944
PCOS (n=3)	α_2-AP	0.051	0.987
PCOS (n=3)	FVII	0.029	0.999
PCOS (n=3)	Fbgn	0.043	0.992
PCOS (n=3)	*Plgn*	*0.153*	*0.615*
Developed PE (n=5)	FVIII	0.097	0.932
Developed PE (n=5)	vWF	0.010	0.986
Developed PE (n=5)	*FIX*	*0.108*	*0.780*
Developed PE (n=5)	FV	0.006	0.987
Developed PE (n=5)	α_2-AP	0.062	0.950
Developed PE (n=5)	FVII	0.045	0.993
Developed PE (n=5)	Fbgn	0.048	0.996
Developed PE (n=5)	*Plgn*	*0.093*	*0.814*

Median Frac. $p > 0.05$ across all 24 pairs: 0.986. Failing pairs (< 0.90, italicized): PCOS–FIX, PCOS–Plgn, DPE–FIX, DPE–Plgn.

Table S11: **Mechanistic validation diagnostics per feature and condition.** Cloud overlap is the fraction of synthetic ODE-predicted/measured ratios falling within the 5th–95th percentile of the corresponding real distribution. KS D and KS p are from a two-sample Kolmogorov–Smirnov test on the same ratio distributions. Real n=69 (23 patients × 3 visits); synthetic n=300 (100 patients × 3 visits).

Condition	TGA Feature	Cloud overlap	KS D	KS p
TF-only	Lagtime	0.92	0.112	0.48
TF-only	Peak	0.91	0.081	0.86
TF-only	T.Peak	0.92	0.102	0.61
TF-only	Max Rate	0.91	0.083	0.84
TF-only	ETP	0.83	0.123	0.36
TF+TM	Lagtime	0.92	0.127	0.32
TF+TM	Peak	0.88	0.079	0.88
TF+TM	T.Peak	0.91	0.110	0.51
TF+TM	Max Rate	0.88	0.096	0.67
TF+TM	ETP	0.89	0.112	0.48

Under TF-only conditions, the cloud overlap range was 0.83–0.92. Under TF+TM conditions, the range was 0.88–0.92. Pooled across both conditions and all five features, the overlap was 0.902. KS D range across both conditions: 0.079–0.127, all $p > 0.30$.

Table S12: **Downstream utility — per-feature held-out V2/V3 performance.** Median relative error of the BZ2012 ODE-predicted TGA values against the held-out real V2 and V3 measurements, for the model calibrated on real V1 patients (K=23) and the model calibrated on synthetic V1 patients (N=100). The synth-calibrated model achieves slightly lower error on every feature. Synth/Real ratio < 1 favors the synth-calibrated model.

TGA Feature	Real-cal MRE	Synth-cal MRE	Synth/Real
Lagtime	0.165	0.162	0.98×
Peak	0.097	0.087	0.90×
T.Peak	0.330	0.306	0.93×
Max Rate	0.286	0.259	0.91×
ETP	0.197	0.183	0.93×
Overall median	**0.201**	**0.189**	**0.94×**

Table S13: **PCA-space ablation.** Five generators operating in the identical d=18 PCA subspace and evaluated on the same harness. Median relative error (MRE) measures marginal fidelity. Mechanistic cloud overlap was the fraction of predicted-to-recorded TGA ratios for synthetic patients within the real-data 5th–95th percentile range after pooling across patients, visits, and TGA features under TF-only conditions. Median novelty is $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \hat{\mathbf{m}}_k)$, and Frac. novel is the fraction above the operational 0.2 cutoff, equivalent to cosine similarity below 0.8 to every stored pattern. The cross-visit Frobenius error is the deviation of the synthetic correlation matrix from the real one. A lower value indicates closer in-sample reproduction and can therefore reward memorization. The comparison showed that the shared representation accounted for much of the marginal and mechanistic fidelity but did not guarantee performance.

Method	MRE (%)	Cross-visit Frob.	Mech. overlap	Median novelty	Frac. novel
SA	1.24	0.641	0.902	0.514	1.00
PCA+GMM	1.13	0.422	0.918	0.497	0.94
PCA+KDE	1.94	0.337	0.823	0.249	0.65
PCA+kNN	1.63	0.587	0.938	0.051	0.05
PCA+Copula	1.34	0.417	0.934	0.462	1.00

Table S14: **Convex-hull membership did not discriminate at K=23 in d=18.** Hull membership and distance for the 100 synthetic patients against the 23 real patients, and for each real patient against the other 22. The radial factor t^* is where the ray from the hull centroid through a point exits the hull, so $1/t^*$ is the factor by which a point overshoots the boundary and $t^* \geq 1$ would mean the point is inside. Because binary hull membership did not show how far a point lay beyond the boundary, we also compared distance to the hull. The real-patient calculation omitted the patient being evaluated. To make the synthetic-patient calculation equally stringent, we omitted each synthetic patient's nearest real patient, leaving 22 real points in both comparisons. A descriptive Mann–Whitney test did not detect a difference in distance between synthetic and real patients ($p = 0.07$), while radial overshoot was smaller for the synthetic patients ($p < 10^{-4}$). Both tests treated the 100 synthetic patients as independent, which they were not, so the reported significance was optimistic. With only 23 patients in 18 dimensions, every real patient formed an outer corner and the hull occupied little of the surrounding space. The generator also normalized the stored patterns and generated directions to length one. This meant that a generated direction could remain inside the hull only by exactly matching a stored pattern.

The biological constraint included non-negativity of both hormones. Negative values occurred for estradiol in 19 patients and progesterone in 37 patients, with 10 overlapping; most were close to zero relative to the real-cohort scale, but the minima were −2109.8 and −75.8, respectively. All nine coagulation-factor inputs to the BZ2012 model were positive, so all 100 synthetic patients were retained for the mechanistic check. Gaussian sampling in PCA space followed by linear decoding had unbounded support, and no clipping was applied. The unconstrained concatenated MVN baseline produced 58 negative hormone measurements across 43 of 100 patients, compared with 57 measurements across 46 of 100 for SA.

Quantity	Synthetic (n=100)	Real, leave-one-out (n=23)
Fraction inside hull	0.00	0.00
Median distance to hull	11.0	11.9
Median radial overshoot $1/t^*$	10.5	15.2
Hull geometry of the real cohort		
Median patient radius	13.8	
Median hull extent, generic direction	2.0	
Largest t^* attained on the unit sphere	0.231	
Plausibility of the out-of-hull synthetic patients		
Biological constraints satisfied	54/100	
Mechanistic band, per feature	83 to 92%	
Mechanistic band, all features and visits	41/100	
Both biological and strict mechanistic criteria	24/100	

Table S15: **Membership inference recovered memory membership, not unseen patients.** Median nearest-neighbor distances in the standardized 216-dimensional space, and the test's discrimination. For each split, we constructed the memory matrix from 15 patients and scored each real record by its distance to the resulting synthetic set, treating the 15 included patients as members and the 8 held-out patients as non-members. Every quantity came from the same protocol. We drew 40 partitions at random with the generation seed held fixed, so that only the partition varied, and pooled the distances across all partitions. Non-members sat at the same distance from the synthetic cohort as unrelated real patients sat from one another. The synthetic cohort carried no proximity signal about patients the generator never saw, and the discrimination came entirely from members sitting closer. The AUC measured how often the test ranked a member above a non-member; 0.5 indicated no discrimination and 1.0 indicated perfect discrimination.

Quantity	Value
Median nearest-neighbor distance	
Memory-matrix member to nearest synthetic	10.7
Held-out non-member to nearest synthetic	15.4
Real to nearest real, leave-one-out	15.9
Synthetic to nearest real	13.8
Membership-inference AUC over 40 *random splits*	
Mean	0.97
Standard deviation	0.02
Median	0.98
Minimum	0.90
Maximum	1.00
Splits reaching perfect separation	9/40
Splits with AUC above 0.9	39/40

Supplementary Figures

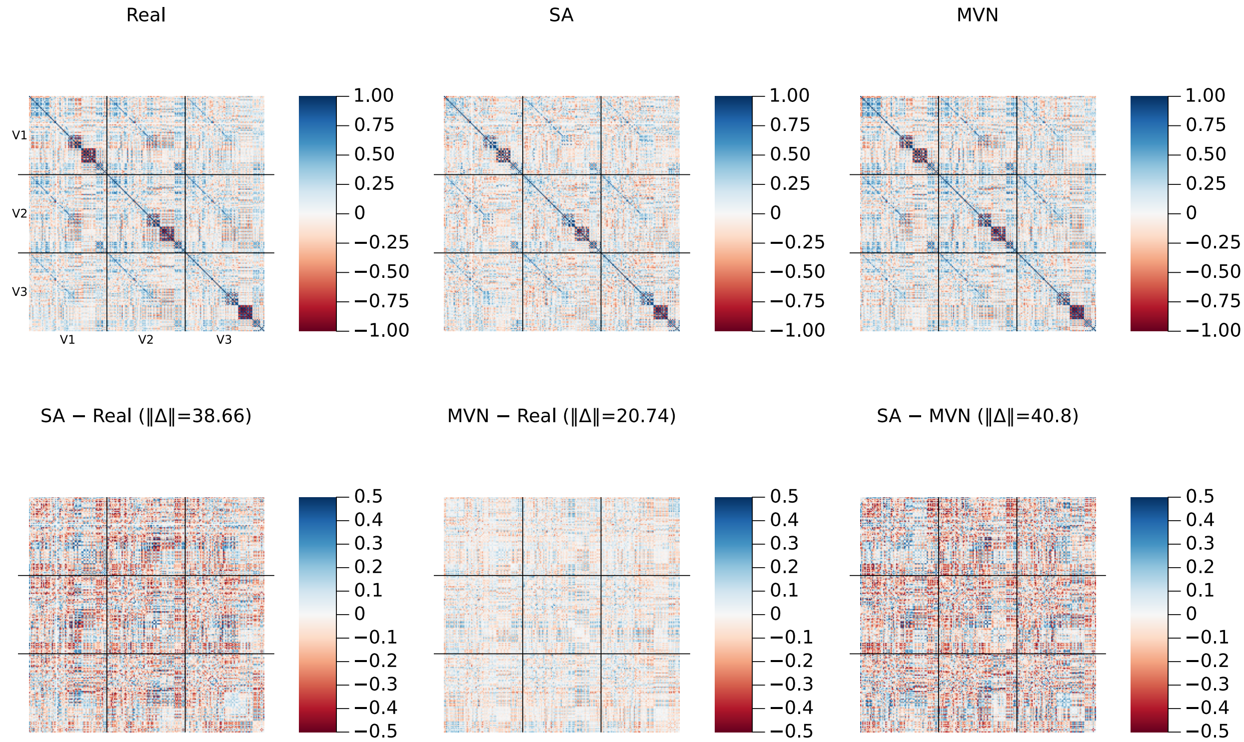

Figure S1: **Cross-visit correlation structure (amplified residual scale).** Same layout as the main-text correlation figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure. Frobenius norms of each residual matrix are shown in the panel titles.

Supplementary Methods: locating the critical inverse temperature

The operating point β^* was located from the attention entropy. For a candidate β we computed the Shannon entropy of the attention weights $\mathrm{softmax}(\beta\,\hat{\mathbf{X}}^\top \boldsymbol{\xi})$ at 20 probe points placed at stored patterns, averaged those entropies, and normalized by $\log K$. The normalized curve runs from 0 to 1. A value of 0 means one pattern carries all the weight, and a value of 1 means every pattern carries equal weight. Sweeping β logarithmically over 80 values from 0.1 to 1000 traces this curve from the diffuse regime into the concentrated one. We took β^* to be the point of most negative second derivative of the normalized entropy with respect to $\log \beta$, computed by finite differences over the sweep. That point is the sharpest bend in the curve, and it marks where attention stops being shared across patterns and begins to localize on one. For multiplicity-weighted sampling we repeated the same computation with the $\log r_k$ bias included in the softmax logits. This gives a ρ-dependent $\beta^*(\rho)$, because the weighting changes how quickly attention concentrates as β grows. For random unit-norm patterns the transition is predicted at $\beta^* = \sqrt{d}$, the theoretical reference quoted in the main text.

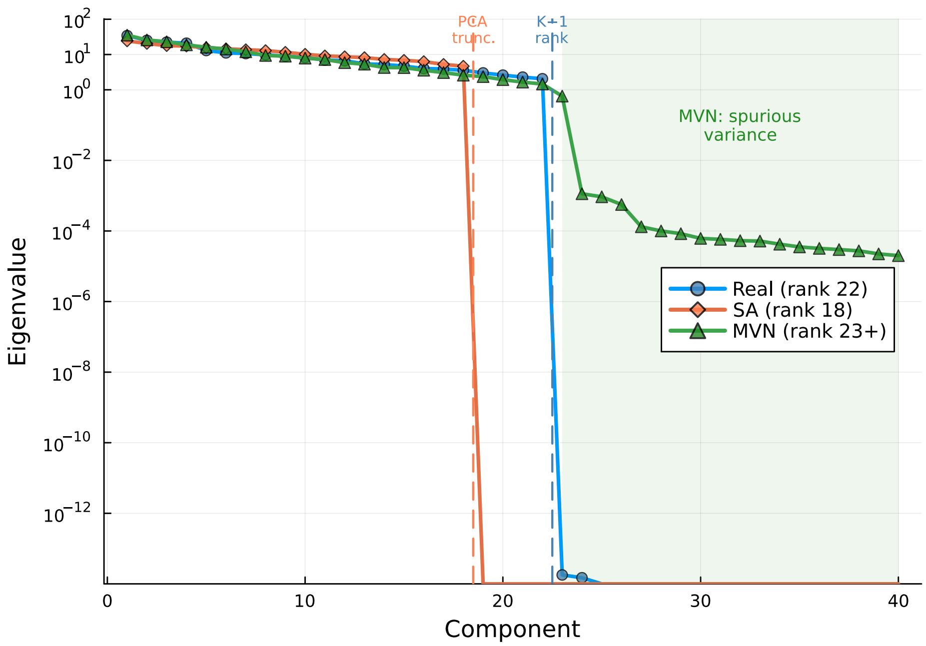

Figure S2: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216×216 sample covariance for Real (K=23, rank 22), SA (N=100, rank 18), and MVN (N=100) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22). MVN's Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain no signal.

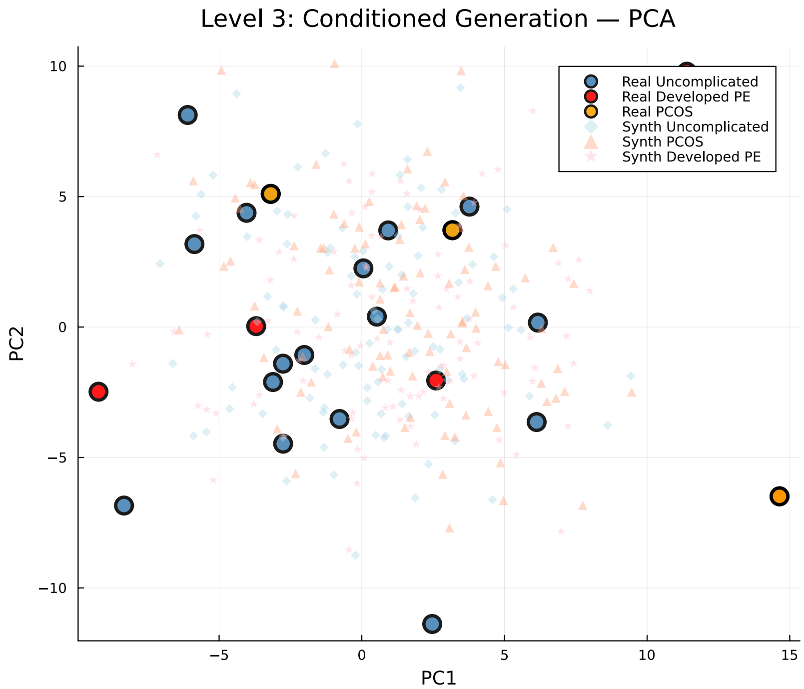

Figure S3: **Conditioned generation: PCA projection (pooled).** Large markers: real patients (blue = Uncomplicated, n=18; orange = PCOS, n=3; red = Developed PE, n=5). Small markers: synthetic patients (N=100 per condition). With only 3 PCOS and 5 Developed PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

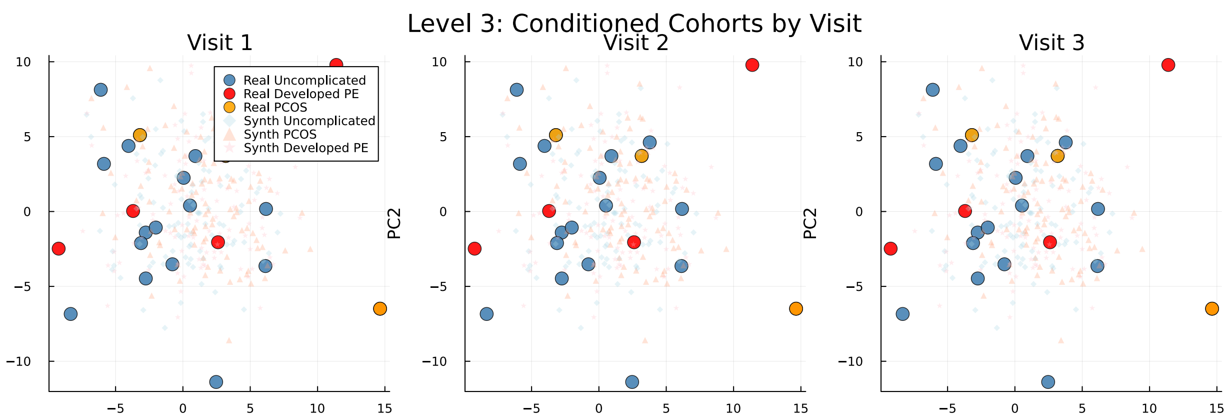

Figure S4: **Conditioned generation: PCA projection by visit.** Same as Fig. S3 but separated by visit (V1, V2, V3). Condition-specific clustering is maintained within each individual visit.

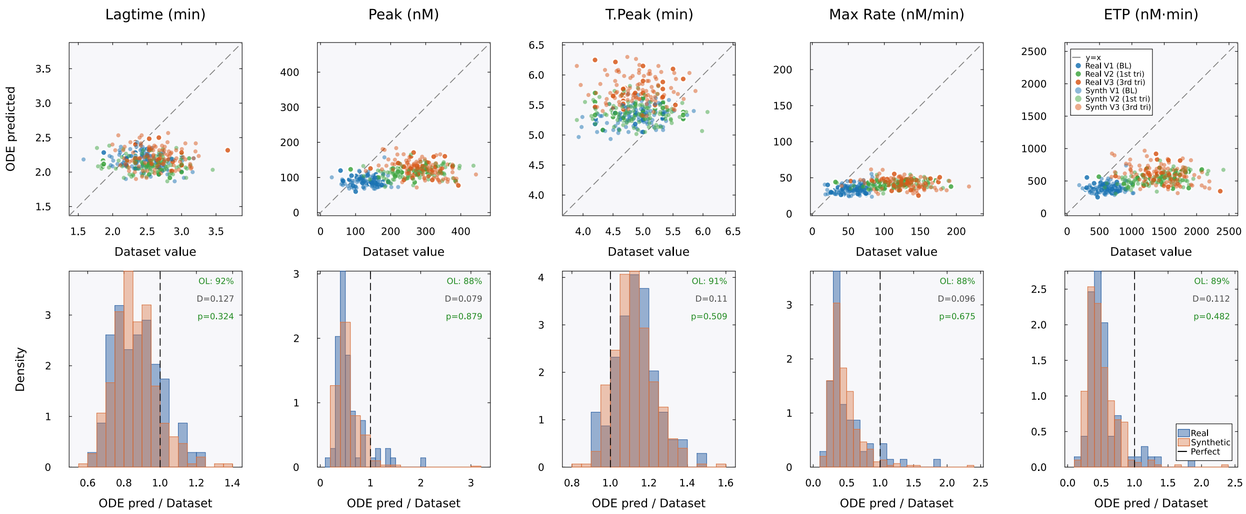

Figure S5: **Mechanistic validation (TF+TM condition).** Same layout as main-text Fig. 6 but under TF+TM conditions. The model systematically underpredicts peak and maximum rate ($\approx 0.5\times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. Cloud overlap remains high at 88–92%, confirming that the model processes real and synthetic patients identically even under imperfect calibration conditions.

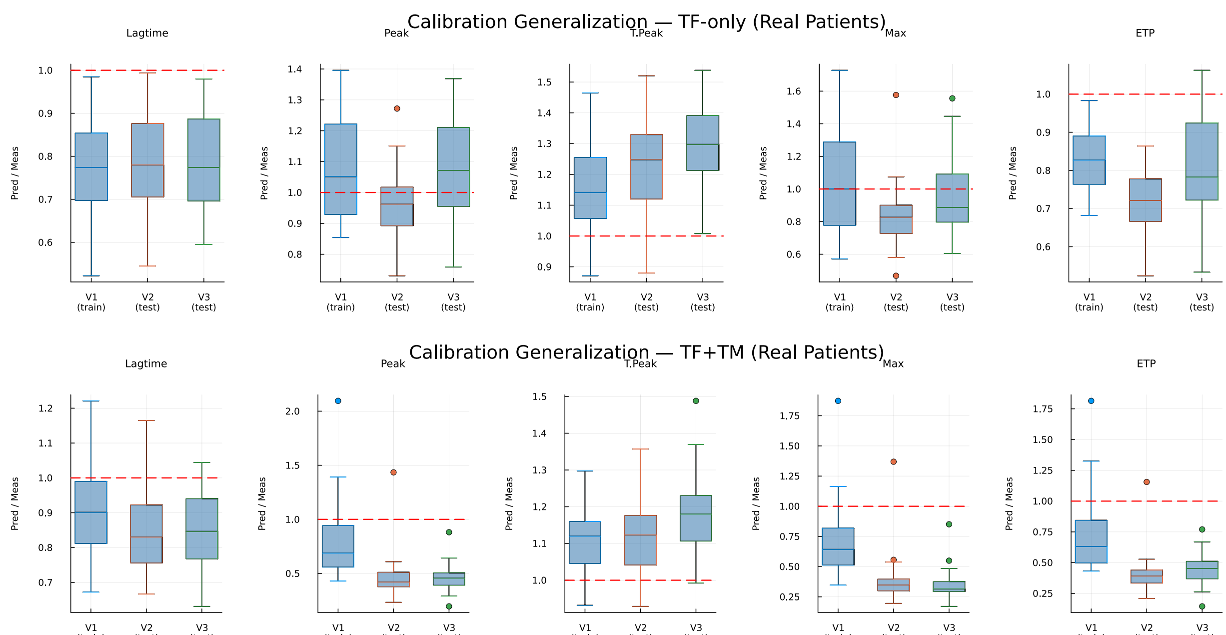

Figure S6: **Cross-visit calibration generalization.** Boxplots of the ODE-predicted/dataset ratio by visit for real patients. The model was calibrated on Visit 1 only. Top: TF-only conditions, where ratios near 1.0 at Visits 2 and 3 indicate that the calibration generalizes across pregnancy timepoints. Bottom: TF+TM conditions, where the calibration does not generalize as well, particularly for peak and maximum rate.

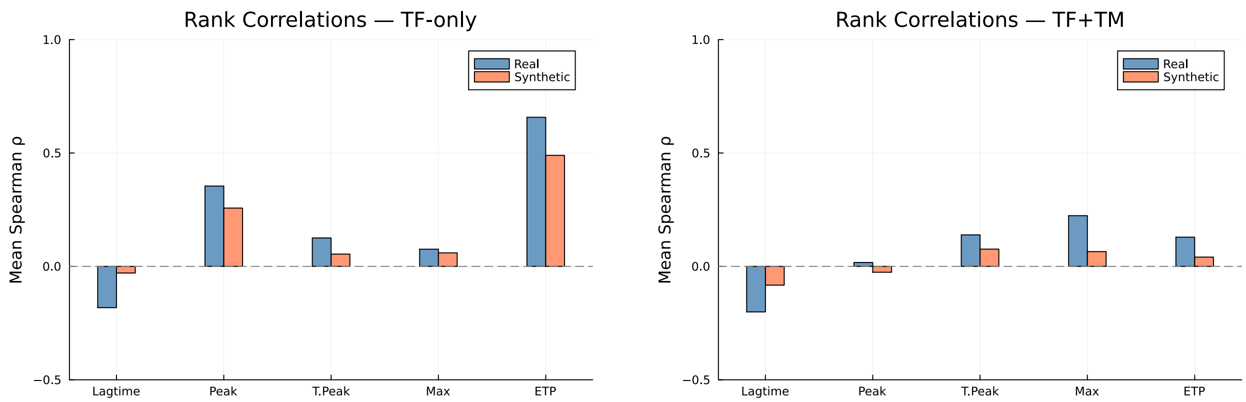

Figure S7: **Spearman rank correlations between ODE-predicted and dataset TGA features.** Left: TF-only. Right: TF+TM. ETP is the best-predicted feature ($\rho \approx 0.6$–0.8 for real patients under TF-only). Weak rank correlations for other features are expected given the 5-parameter population-level calibration.

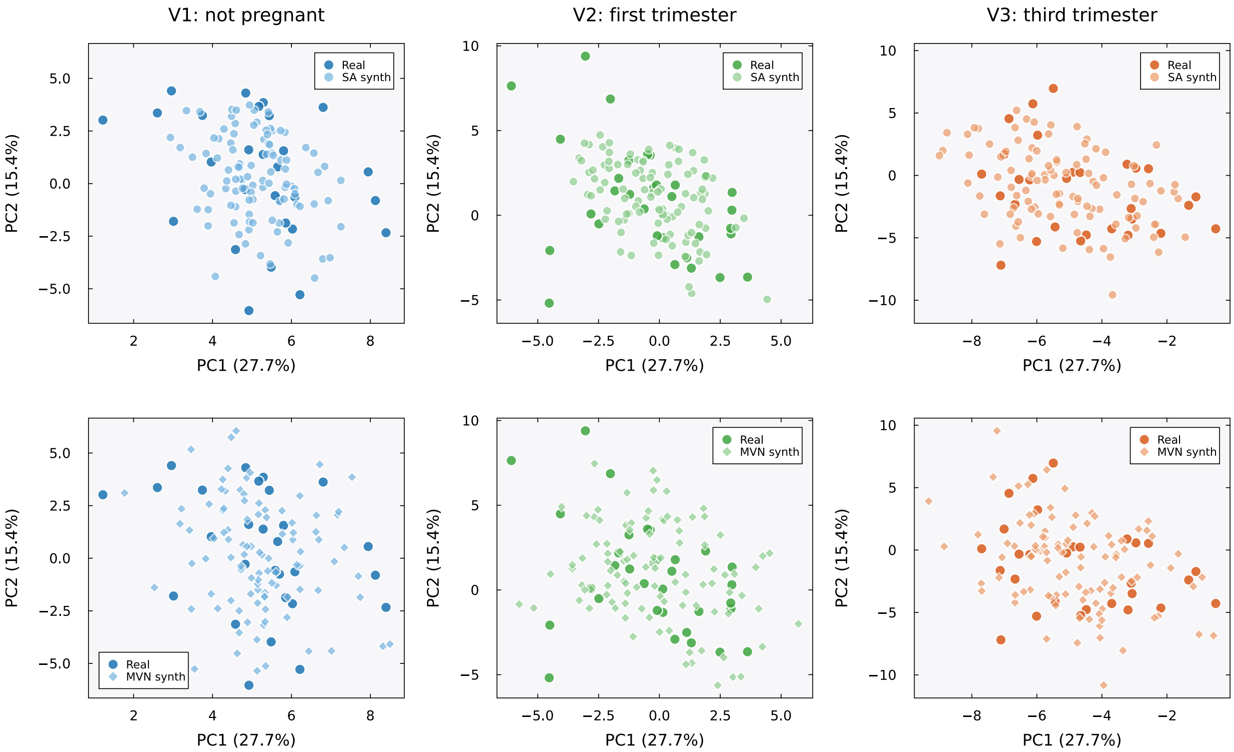

Figure S8: **PCA projections by visit: SA vs. MVN.** Each panel shows the first two principal components (PC1: 27.7% variance, PC2: 15.4%) computed from standardized per-visit real patient data (K=23 patients per visit, 72 features). Dark markers indicate real patients; lighter markers indicate synthetic patients (N=100). Within each visit (column), the SA panel (top) and MVN panel (bottom) share identical axis ranges, so the relative dispersion of the two methods can be read against the same real data cloud. Top row: SA-generated patients (circles) cluster tightly around the real data cloud at all three visits. Bottom row: MVN-generated patients (diamonds) show greater dispersion, particularly at Visits 2 (first trimester) and 3 (third trimester), where MVN patients extend beyond the convex hull of the real data. This excess scatter reflects the spurious variance introduced by Ledoit–Wolf regularization of the rank-deficient covariance (Fig. S2). Colors encode visit: blue (V1, not pregnant), green (V2, first trimester), orange (V3, third trimester).

Sampling correctness and privacy diagnostics

We assessed a narrowly defined membership-inference question: if someone already possessed a candidate de-identified assay profile, could the synthetic cohort reveal whether that record had been included in the memory matrix? The source cohort contained no direct identifiers or link to named individuals, so this was not a test of personal identification. For every synthetic patient we computed the distance to the closest real record (DCR) in the standardized 216-dimensional space and compared the synthetic-to-real distribution against the leave-one-out real-to-real distribution. Synthetic patients sat slightly closer to the real cohort than the real patients sat to one another, with median DCR 13.8 against 15.9. We also ran a nearest-neighbor membership-inference test, which scored each real record by its proximity to the synthetic set over repeated random train/holdout splits. It separated the 15 memory-matrix members from the 8 held-out patients with a mean AUC of 0.97 (standard deviation 0.02 over 40 splits; 0.5 indicated no discrimination and 1.0 perfect discrimination). At the published operating point the generator therefore retained enough record-specific proximity for memory-matrix membership to be detected.

To attribute this behavior we asked whether it reflected the target distribution $\pi(\xi) \propto \exp(-\beta E)$ or merely the sampler used to draw from it. A mixing diagnostic showed that at β^* the target was multimodal, with one component centered on each of the K=23 stored patterns. It also showed that discretization bias in the Langevin updates was negligible, with Metropolis acceptance near 0.999 (Supplementary Fig. S10). We then constructed a four-rung ladder (Table S16, Fig. S9) that held β, the memory, and the decoding fixed and varied only the sampler. The rungs ran from the published anchor-initialized single-endpoint scheme, to sphere initialization, to burn-in-controlled thinned pooling, to Metropolis-adjusted sampling. Across the ladder the membership-inference AUC stayed near 0.96 to 0.97. The median DCR rose only from 14.0 to 14.3 and never reached the real-to-real baseline of 15.9, at a modest cost in marginal fidelity. Proper sampling of π did not remove the memorization. At K=23 the memorization was a property of the distribution.

The remaining lever was the inverse temperature. Lowering β spread probability mass away from the stored patterns and was expected to improve privacy, at the expense of the memorization that also underlay fidelity. We swept β across a 40-fold range around β^* (Fig. S9C). The sweep confirmed the trade-off and bounded it sharply. The median DCR rose monotonically as β fell, but it saturated near 14.3 and stayed below the real-to-real baseline of 15.9 even at one twentieth of β^*. Over the same range the cross-visit covariance error grew from 0.59 to 0.69. Marginal and mechanistic fidelity were insensitive to β, so the longitudinal covariance structure bore the cost of lowering it. No inverse temperature yielded a cohort that was both as private as the real patients were from one another and faithful to their joint structure.

Neither lever settled the question completely, because every scheme compared was a finite-time Markov chain. The target itself was available in closed form. Completing the square in Eq. (1) gives Eq. (3). For unit-normalized memories π_r is exactly the isotropic Gaussian mixture $\sum_k q_k \mathcal{N}(\mathbf{m}_k, \beta^{-1}\mathbf{I})$ with $q_k = r_k / \sum_j r_j$. It can therefore be sampled ancestrally, by drawing a component and adding isotropic noise. There is no chain, no initialization, no burn-in and no discretization error. We generated a protocol-matched exact cohort (N=100 at β^*, reusing the ladder's seeded magnitude draw so that only the direction sampler differs) and evaluated it on the identical harness (Table S16). Every metric fell inside the range spanned by the four rungs: median relative error 1.68%, cross-visit Frobenius error 0.60, mechanistic overlap 0.91, median novelty 0.50, and median DCR 13.87. One hundred independently seeded exact cohorts placed these quantities in narrow intervals, with median relative error 1.40% (fifth to ninety-fifth percentile 0.97 to 1.91%) and median DCR 13.95 (13.61 to 14.39). The agreement did not rest on a favorable seed. Over the same 40 train/holdout splits used for the ladder endpoints, the membership-inference AUC under exact sampling was 0.98, marginally above the published scheme's 0.97. The memorization was therefore a property of π_r at K=23, where every stored patient was a component center. It was not a property of any approximation used to sample it.

Table S16: **Sampler-correctness comparison.** Each rung held β^*, the memory, and the decoding fixed and varied only the direction sampler. Fidelity metrics (median relative error, cross-visit correlation Frobenius error, mechanistic cloud overlap, median novelty) and privacy metrics (median distance to the closest real record, and repeated-split membership-inference AUC where evaluated) were reported per rung. The real-to-real DCR baseline was 15.9. The membership-inference test was run on the two endpoints of the ladder and on the analytic reference, over the same 40 train/holdout splits. *Exact* was not a rung. It drew ancestrally from Eq. (3) with no chain, and served as the analytic reference for the four sampled rungs. Over 100 independently seeded exact cohorts the median relative error was 1.40% (fifth to ninety-fifth percentile 0.97 to 1.91%) and the median DCR was 13.95 (13.61 to 14.39).

Rung	Sampler	MRE (%)	Cross-visit Frob.	Mech. overlap	Novelty	DCR	MIA AUC
V0	anchor, endpoint	1.31	0.65	0.89	0.51	13.97	0.97
V1	sphere init	1.53	0.57	0.89	0.50	13.96	n/a
V2	pooled ULA	2.02	0.59	0.88	0.50	14.15	n/a
V3	pooled MALA	1.89	0.62	0.92	0.52	14.33	0.96
Exact	analytic ancestral	1.68	0.60	0.91	0.50	13.87	0.98

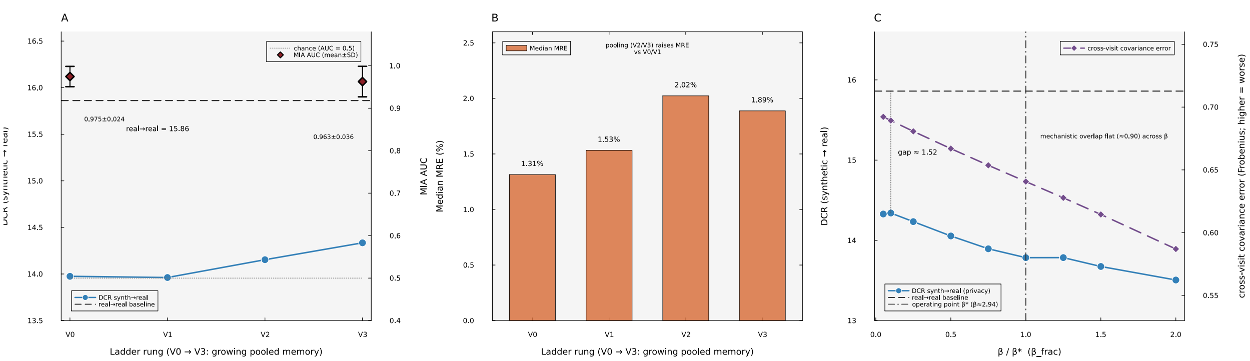

Figure S9: **Sampling correctness and the memorization/privacy relationship.** (A) Across the V0 to V3 ladder the distance to the closest real record (DCR, left axis) rose only slightly and stayed below the real-to-real baseline, while the membership-inference AUC (right axis) remained near 0.96 to 0.97. (B) The fidelity cost of multi-chain pooling, as median relative error. (C) The inverse temperature swept across a 40-fold range around the operating point β^*. The DCR (privacy, left axis) rose as β fell but never reached the real-to-real baseline. The cross-visit covariance error (right axis) grew, and the mechanistic overlap stayed flat.

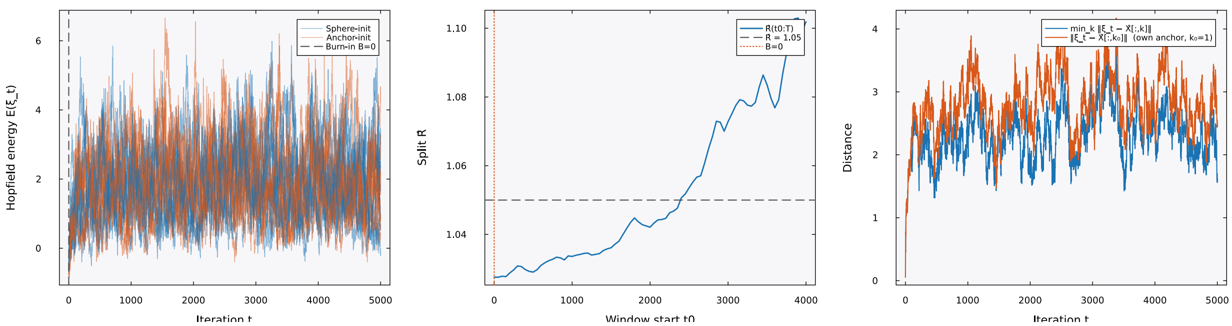

Figure S10: **Mixing diagnostic at β^*.** Energy traces, the between-chain $\hat{R}$ statistic, and the anchor-distance trajectory for sphere-initialized Langevin chains. The target distribution at β^* was multimodal, with one component centered on each stored pattern. Metropolis acceptance near 0.999 indicated negligible discretization bias.